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Improving pressure monitoring and control in order to reduce water loss in water urban public systems

To cite this article: F Stătescu *et al* 2024 *IOP Conf. Ser.: Mater. Sci. Eng.* **1304** 012031

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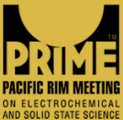
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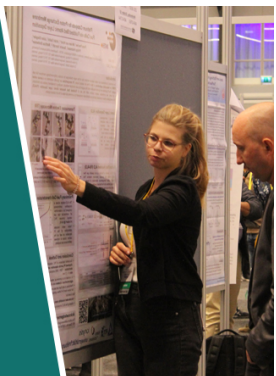
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Improving pressure monitoring and control in order to reduce water loss in water urban public systems

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Abstract. Water losses from distribution systems are present in all situations. Reducing them to zero is impossible. Modern technologies for operating water distribution networks are an effective means of reducing water losses. In this sense, the conducted research aimed at testing some solutions to improve the monitoring and control of pressure in a distribution network. The mathematical model of the water distribution network was developed and calibrated (Mike Urban application) and the hydraulic modeling of the network was carried out. The results were obtained through: periodic simulations with the help of the monitoring application; comparisons for flow and pressure values, data collected from field measurements, data obtained from SCADA application. These complex analyzes allow the identification, in the shortest possible time, of existing problems and areas with losses.

1. Introduction

Drinking water is often considered an element of nature that we have access to in a natural way, just as we benefit from the air we breathe. We lose sight of the fact that drinking water is a product, and its production in a sufficient quantity is achieved through a complex process [1]. If we also consider the inadequate management of water distribution, which is present in many situations, a less visible but crucial problem is highlighted, that of water losses, part of non-revenue water (NRW).

Reducing all components of water loss to zero is neither technically possible nor economically viable. Components of water loss must be accurately assessed and prioritized for the reduction process.

Operating a drinking water distribution network is no easy task. Such a network serves a large area, consisting of a complex system of pipes, equipment and accessories. For these reasons, it is very difficult to detect damages, modifications and illegal connections.

Water losses are caused by multiple causes: broken pipes and water leaks, poor water management, illegal connections and unauthorized consumption. Fortunately, just as varied are the solutions that can be applied with considerable benefit.

Pipe breaks and water leaks affect both the water company and the environment. This is due to the fact that lost drinking water no longer reaches consumers and also negatively affects the environment. These pipe breaks and water leaks can occur suddenly or gradually as a result of corrosion, wear and tear or lack of maintenance.

Mismanagement of drinking water leads to the impossibility of billing by water companies for the actual consumption. Competent and appropriate management considers the permanent maintenance of water distribution networks in order to ensure normal operating parameters and permanent checks on the customers served.

Illegal connections and unauthorized consumption can in some cases represent a substantial percentage of water losses in distribution networks.

2. Reducing water losses through advanced strategies and technologies

Losses in drinking water distribution networks can be reduced by several methods [2,3,4]:



- Dividing the distribution network into districts. A technique for an efficient operation of drinking water distribution networks is to divide it into sectors, also called district metering areas (DMA). Thus, water losses can be assessed for each sector, and operators can plan and prioritize their actions.
- Rapid damage assessment and repair. Quick actions lead to saving resources, protecting the environment and causing less inconvenience to customers who need a continuous supply of water.
- Monitoring network activities. Different techniques can be applied to measure and manage water pressure at different points in the distribution network.
- Pressure management. This is the most effective method of managing potable water losses. Most pipe breaks occur not so much due to high pressures, but mainly due to pressure fluctuations, which cause continuous expansion and contraction of pipes, resulting in their rupture. The higher the network pressures, the higher the amount of water losses.
- Collection of data on network operation and taking the correct decisions. The data collected in real time is an essential support in making sound decisions regarding the reduction of water losses.
- Fighting illegal consumption. There are effective practical ways to detect and monitor illegal drainage and water theft.
- Use of quality products and solutions. Any effective drinking water distribution system is based on the use of high-quality resources and solutions. The research carried out on the „automated installation for the verification of flow transducers and compact and combined thermal energy meters, Code MY SMIS: 122085”, from the Faculty of Hydrotechnics, Geodesy and Environmental Engineering, Technical University "Gh. Asachi" Technical of Iasi, can offer modern and efficient solutions for the optimal exploitation of water distribution networks [5].
- High level vocational training and education. Scientific research, knowledge sharing and access to good techniques are essential to ensure that the water supply sector is able to meet future challenges.

3. Research performed on the zonal distribution system: Iași Municipality and the Metropolitan Area

3.1. System characteristics and the need for pressure control

The distribution system of Iași City (Fig. 1) has the peculiarity of the need to pump water in stages, from ground level +40 to ground level +400. Hence, this system includes a series of intermediate pumping stations, storage tanks, compensation devices for the maximum hourly consumption and a series of local booster pumping stations.



Figure 1. Location of water distribution system: Iași City and the Iași Metropolitan Zone.

The main constructive elements of the system are shown in Table 1 and Fig.2.

Table 1. Constructive features of Iași City and Iași Metropolitan Zone water systems.

Year	Length (km)	Pumping stations (pcs)	Re-chlorination station (pcs)	Tank		Connections	
				No	Storage volume	Population	Economic agents
				(pcs)	(m ³)	(pcs)	(pcs)
2022	1426	38	17	61	118.825	49.280	6215

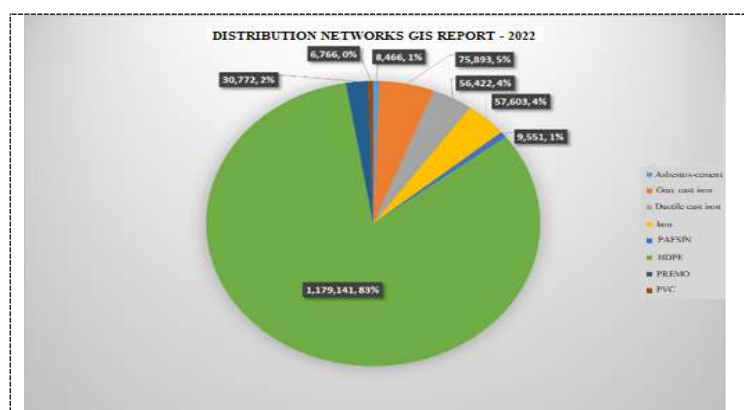


Figure 2. Pipes Materials: asbestos-cement / gray cast iron / ductile cast iron / steel.

The Iasi system, like most water supply systems, is designed to ensure a minimum working pressure at all points of the system throughout the day. It therefore represents the minimum pressure at a critical point in the system, which is often either the highest point in the system or the furthest from the source.

The system features significant fluctuations in water demand throughout the day, with peaks in the morning and evening, along with periods of low demand at night and sometimes in the early afternoon. In many residential areas, especially the newly established ones, there are also seasonal fluctuations, caused by climatic factors with an impact on irrigation needs or migration during vacations, for periods of days or weeks.

In accordance with applicable design standards, the water supply system is designed to meet the pressure requirement during peak demand periods when head losses are at their highest and internal pressures are at their lowest. As a result of this design methodology, the system experiences higher pressures than necessary during the off-peak period. This is clear from the fact that in most areas, major breakdowns occur in the late evening and early morning when system pressures are the highest in order to meet water demand.

The pressures at which the distribution network operates must be rigorously monitored and controlled, in order to reach the optimal level, which ensures the delivery of water to consumers in a sufficient and efficient manner, according to the supply contract. The benefits of this process are reducing the number of breakdowns due to high static/transient pressure and thus, reducing the magnitude of actual losses. However, a high-performance design and operation of the network is necessary to ensure consumers the requested and contracted level of service. Active pressure control has as its main objective the reduction of excessive pressure in a water distribution system, which results in the reduction of water losses and implicitly the frequency of failures in the network.

3.2. *Work methodology*

3.2.1. *Research directions* In order to improve pressure monitoring and control, the mathematical model of the water distribution network (Mike Urban application) was developed and calibrated, and a network hydraulic modeling was carried out.

The established research plan includes: periodic simulations using the application for monitoring; comparisons for flow and pressure values, resulting from field measurements, with data from the SCADA application. These comparisons allow the identification, in the shortest possible time, of existing problems and loss areas.

The recalibration of the mathematical model of the network is carried out at a time interval of 5 years, or when important structural changes of the network are implemented, which can be:

- expanding the network in neighboring areas;
- alternation of measurement areas on existing neighborhoods;
- new measurement areas by DMAs;
- the construction of new supply/transport pipelines.

3.2.2. *Implementation of new methods for pressure control* The improvement of the monitoring and control of pressure in the system was carried out by implementing a plan of measures, which includes:

- Installation of valves and creating a system zoning, an activity that involves the partial or total closing of the valves in the system, having the effect of reducing the transport capacity in the network, or moving a supply area to a lower pressure. Particular attention was paid to the activity of defining the size of the zones, so that the water demand for fire hydrants and for refilling the system after a major breakdown on a main water pipeline are ensured.
- The control and reduction of the pumping height was applied in the case of areas where the water demand decreased.
- The installation of pressure reducing valves is the most frequently used method of reducing pressure in distribution networks (Fig. 3). Various types of valves have been used, such as: valves that produce a constant pressure between inlet and outlet, valves that provide a constant

downstream pressure, valves that provide a variable downstream pressure allowing pressure drops during the night. Due to their sensitivity in operation, the pressure reducing valves (PRV) are periodically checked and provided with a by pass.



Figure 3. Pressure decreasing device, AVK, Dn 1000 mm, fitted with Pilot timer for a Dn 1000 pressure adjusting valve used for setting two pressure thresholds, including a retrofit (CHA KIT) and an electronic e-Timer-33 module.

- Programming the frequency converters for preset pressure maintenance hourly intervals through which the reduction of the pressure supplied by pumping equipment was achieved in the hourly interval 23:00-03:00, when water consumption was at minimum values.

In order to carry out the research, the following elements were taken into account:

- the number of connections;
- the length of distribution pipes;
- the number of properties;
- the population served and type of consumption: domestic, and non-domestic;
- the probable rates of physical losses;
- the pressure exponent for the system;
- details of any commercial/industrial consumers.

The conveyed data are the same as those used in the minimum night flow (MNF) analysis. In addition, three types of pressure profile and the water demand of the area in 24 hours have been identified. The average hourly values must be known at the following characteristic points:

- the pressure at the entry point (p_{inlet});
- the pressure at the middle point of the zone (p_{center});
- the pressure at the critical point, at the most disadvantaged consumer ($p_{critical}$);
- water demand in the area or DMA ($Q_{hr.med}$).

These four sets of hourly values are continuously measured, using water meters and electromagnetic flowmeters to measure flow and pressure loggers to measure pressure. All measuring equipment data is entered into the SCADA BlueMonitor application, they provide data at 15-minute intervals and are permanently monitored. For critical values, alarms have been set and are automatically transmitted to the operating personnel on field.

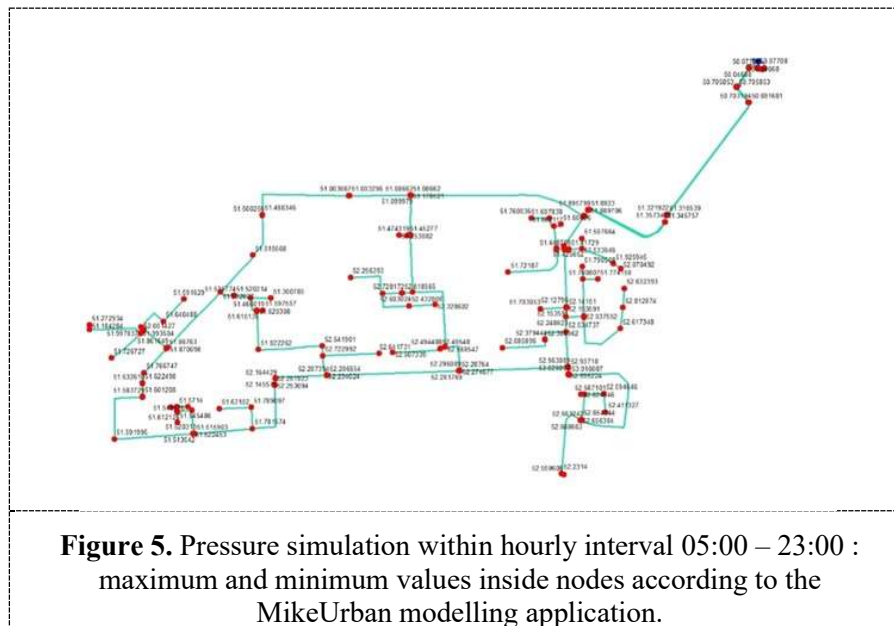
- The complete solution implemented in the distribution system of Aroneanu commune, Iași county (Fig. 4), includes the Pegasus 2 system, with a control unit with bidirectional communication system, MultiLog data-loggers and the PressView software platform. Thanks to the two-way communication system, the equipment can remotely control the pressure regulator valve using the latest GPRS communication technologies: LTE-M, NBIoT, 4G, and 3G with downgrade to 2G.



Figure 4. The Pegasus system, connected to an automatic pressure control valve, a MUT 1222 insertion flow meter and a MultiLog2 data logger (featuring an integrated telemetry system).

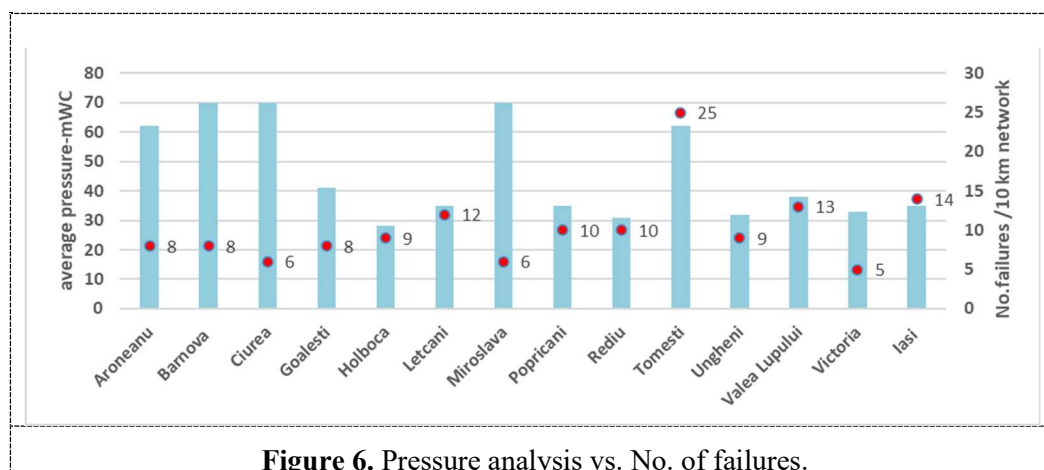
3.3. Obtained results

3.3.1. Simulation of network operation Work was performed under the MikeUrban DHI application, a specialized program, compatible with the program related to the GIS application, so that common data is exported to the modeling program. Mathematical models of the networks were performed for the localities within operating area, which include: the network graph with all the basic data (Fig. 5): flow rate and supply mode (tank, pumping); the method of extracting water from network – directly to consumers or through pumping stations - repumping (characteristics); network graph (bar connection, bar lengths, diameters, material, roughness, topographic elevations at nodes).



In order to calibrate the hydraulic model, flow/pressure measurements were performed at predetermined points, such as: supply nodes, storage points and critical points. The model thus calibrated can simulate any section of a day or period of time. It offers the possibility of studying the behavior of the network in various operating situations, the scenarios being adapted in direct relation to the operational needs.

3.3.2. Comparative analysis: pressure vs. number of failures The research on this issue was carried out in several localities, and the results are presented in Fig. 6. It was studied whether there is a connection between the minimum pressure in the network and the number of failures, the study being made for 10 Km of network. It can be seen that a general relationship cannot be defined only between these two parameters. The causes that determine the occurrence of failures are much more numerous.



Collected data were processed in GIS ArcMap, and hence thematic maps were generated (Fig. 7), maps very useful for the network operation process.

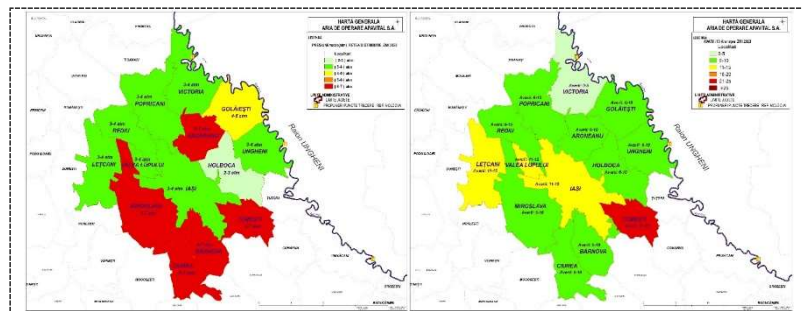
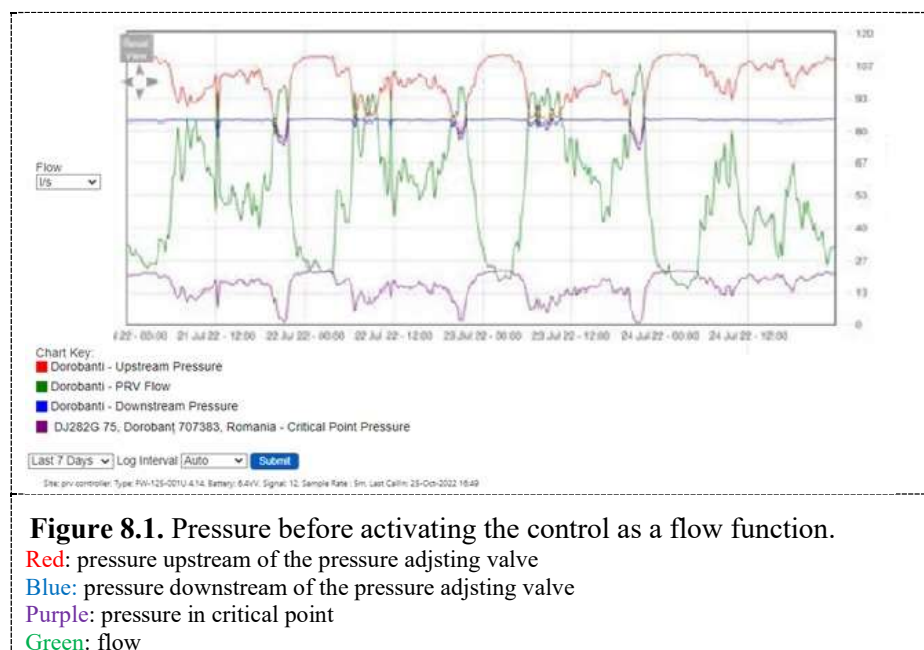


Figure 7. Thematic maps (average pressure, number of failures).

3.3.3. Behavior of the automatic pressure controlling devices The tested solution included: the Pegasus 2 system, with a control unit, cu featuring a bidirectional communication system, the MultiLog data loggers and the PressView platform. The results of a test for this system are shown in Fig. 8.1 and Fig. 8.2.

a) Graph: the pressure before activating the control as a flow function



b) Graph: the pressure after activating the control as a flow function



A good pressure control as a function of flow has been provided, after the activation of this system.

4. Conclusions

The pressure management is a complex process by means of which the pressures at which a distribution network operates are monitored and controlled in order to ensure sufficient and efficient water delivery to consumers. The benefits of this process are: reducing the number of network failures due to high static/transient pressure and reducing the amount of real losses. However, the importance of design and other operating conditions in ensuring the required service level should not be underestimated.

The measurement of pressure and volume must be carried out with metrologically verified measuring instruments, which ensure confidence in the measured values. Determining measurement errors is a mandatory step in testing the functionality of water distribution systems.

Accurate metering of transited water volumes is an essential measure of water loss reduction plans.

Advanced scientific research is a guarantee for the correct and efficient solution of current and future challenges in the field of water distribution.

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competitiveness and business development, Action: POC/163/1/3/Stimulation the demand of companies for innovation through CDI projects carried out by companies individually or in partnership with CD institutes and universities, in order to innovate processes and products in economic sectors with growth potential